Alexandria University
Faculty of Engineering
Computer and Systems Engineering Department

CSE343 Embedded Systems
Assigned: Sunday 11/05/2026
Due: Thursday 21/05/2026

Lab Experiment 08: Machine Learning Inference on Device

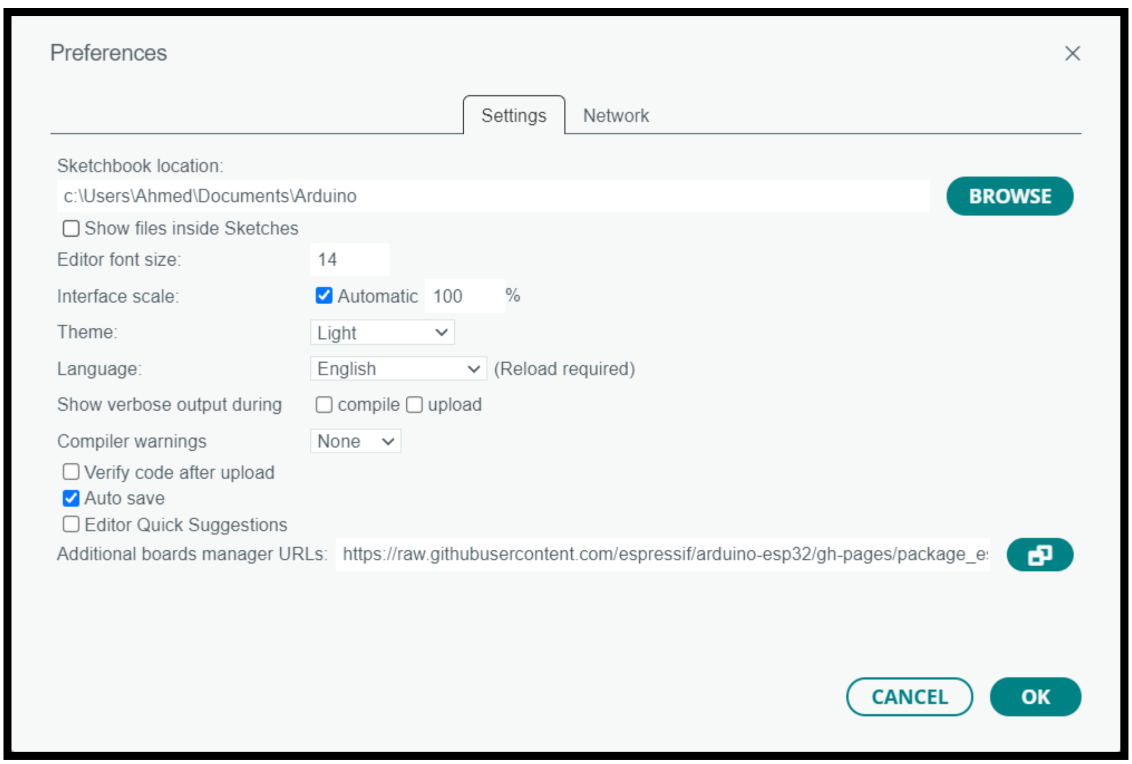
Objectives

- Applying machine learning inference on device (ESP32).

Hint: You can download the project repository:
https://drive.google.com/drive/folders/1awZcYaSKzUFLn3vXe2FMFLsQMtCTMmGy?usp=share_link

Procedure

- Update Arduino IDE to version 2.1.0.
- Add the following link if needed: https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json for viewing esp32 downloadable boards to the preferences as shown below:



- Update ESP32 board using boards manager to version 2.0.6 if needed.

- Choose “ESP32 Wrover Module” from available boards.
- Add the following zip files from project repository in arduino libraries: ***TensorFlowLite_ESP32.zip*** and ***kissfft-master.zip***
  You can find how to import libraries in arduino in the following link: https://support.arduino.cc/hc/en-us/articles/5145457742236-Add-libraries-to-Arduino-IDE
- Upload the following sketch to the esp32 board: ***mfcc_test7/mfcc_test7.ino***
  Note: The audio file and the model are already stored in the code section in flash memory.
- Open the serial monitor and review the result.

References

- Getting started with TensorFlow Lite for microcontrollers
  Link: https://www.tensorflow.org/lite/microcontrollers/get_started_low_level
- Converting TensorFlow Lite model, bmp, and wave files to cc arrays.
  Use the following file: ***generate_cc_arrays.py***

Delivery Policy

- Each group must send the inference serial monitor text in the report after run.
- You should submit a report showing the challenges you faced (if any) and the comments you have on results.
- You should cite any additional resources you used.
- Further details for the submission instructions will be posted later on MS Teams.

Good Luck

Please if you have done any work on training the model, inform us.